MIL-A-63187(AR)
25 June 1982
SUPERSEDING
(See 6.4)

MILITARY SPECIFICATION

ADHESIVE, POLYAMIDE EPOXY M753 8 INCH PROJECTILE

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for polyamide epoxy adhesive used in M753, 8 Inch Projectiles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

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STANDARDS

MILITARY

MIL-STD-1188	Commercial Packing of Supplies and Equipment

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.5)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9321334	Adhesive Packet
11828813	Adhesive, Polyamide Epoxy

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D638	Standard Test Method for Tensile Properties of Plastics
ASTM D792	Specific Gravity of Plastics
ASTM D897	Tensile Properties of Adhesive Bonds
ASTM D1002	Strength Properties of Adhesives in Shear by Tension Loading (Metal-to-Metal)

(Applications for copies of ASTM documents should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Materials. The adhesive system shall be in two parts consisting of an epoxy resin base and a polyamide resin curing agent and shall comply with all requirements specified on drawing (dwg) 11828813, with all requirements specified herein and with the requirements of all applicable specifications to the extent specified on the dwg. and herein. The contractor shall submit test samples of materials to the acquiring agency for test and approval in accordance with the requirements of this specification.

3.2 Adhesive system physical properties.

3.2.1 Specific gravity. The specific gravity of the uncured adhesive system when combined in preparation of 40 parts of resin to 60 parts of activator by weight shall be 1.06 to 1.08 at 23° ± 3°C (73° ± 5°F). Testing shall be performed in accordance with 4.5.1.

3.2.2 Physical properties of cured system. The physical properties of the adhesive system when combined in proportion of 40 parts of resin to 60 parts of activator by weight and cured for 7 days at 23° ± 3°C (73° ± 5°F) or 2 hours at 74° ± 3°F (165° ± 5°F) shall be as indicated in Table I. Testing shall be performed in accordance with the applicable method of 4.5.2.

TABLE I. Physical properties of adhesive system.

Properties	Limits Min.	Max.
Tensile strength (psi)	3500	-
Elongation (%)	10	20
Bond strength (psi)	2000	-
Tensile shear (psi)	2600	-

3.3 First article inspection. This specification contains technical provisions for first article inspection (see 4.3). Requirements for submission of first article samples shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as

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specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract or purchase order, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

1. First article inspection (see 4.3)
2. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. When the contract requires first article inspection (see 3.3), the contractor shall submit a first article sample to a location acceptable to the Government (see 6.3) for evaluation in accordance with 4.3.2. The first article sample shall consist of sufficient material(s) (epoxy resin base and polyamide resin acting agent) to perform the required tests and shall have been produced with the same equipment, processes and procedures which will be used during production.

4.3.2 Inspections to be performed. The first article sample submitted in accordance with 4.3.1 shall be subjected to any or all of the requirements of the applicable dwg. requirements and to the tests and inspections specified in the first article inspection table herein.

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FIRST ARTICLE INSPECTION

MIL-A-63187(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH 4.3.2.1	TITLE Adhesive, Polyamide Epoxy		SHEET 1 OF 1		DRAWING NUMBER 11828813
					NEXT HIGHER ASSEMBLY M753 Proj.
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Specific gravity	(a)		3.2.1	4.5.1
	Physical properties	(a)		3.2.2	4.5.2

NOTES: (a) As required to perform the applicable ASTM tests.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.3.3 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Inspection lot. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, an adhesive system inspection lot shall consist of materials from one manufacturer employing not more than one lot or batch of epoxy resin and one lot or batch of polyamid resin activator.

4.4.2 Examinations. See MIL-A-48078 and the quality conformance inspection table herein. Quality conformance inspection shall be performed on the adhesive system at a location acceptable to the Government (see 6.3). If the quality samples fail to meet any of the applicable adhesive system test or inspection requirements, the inspection lot of materials shall be rejected.

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QUALITY CONFORMANCE INSPECTION

MIL-A-63187(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH 4.4.3.1	TITLE Adhesive, Polyamide Epoxy		SHEET 1 OF 1		DRAWING NUMBER 11828813
					NEXT HIGHER ASSEMBLY M753 Proj.
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Specific gravity	(a)		3.2.1	4.5.1
102.	Physical properties	(a)		3.2.2	4.5.2

NOTES: (a) As required to perform the applicable ASTM tests.

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QUALITY CONFORMANCE INSPECTION

MIL-A-63187(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH 4.4.3.2	TITLE Packaging		SHEET 1 OF 1		DRAWING NUMBER See 5.0
					NEXT HIGHER ASSEMBLY N.A.
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Packaging, packing and marking not in accordance with applicable requirements	1/lot		5.0	Visual
102.	Materials not individually packaged when required.	1/lot		5.0	Visual
103.	Marking missing, incorrect or not legible on individual packages, intermediate packages and external containers, as applicable.	1/lot		5.0	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.5 Test methods.

4.5.1 Specific gravity (see 3.2.1). The specific gravity of the uncured adhesive system shall be determined in accordance with ASTM D792.

4.5.2 Physical properties of cured adhesive system (see 3.2.2).

4.5.2.1 Tensile strength and elongation. Tensile strength and elongation shall be tested on test specimens in accordance with ASTM D638 or equivalent subject to the following conditions:

a. Material for test specimens shall be thoroughly mixed (40 ± 0.4 parts by weight of resin to 60 ± 0.6 parts by weight of curing agent) for a maximum period of 5 minutes.

b. Molded test specimens shall be prepared in accordance with ASTM D638, Type I. Suitable means shall be provided to remove specimens from the mold and to assure against entrapment of air or gases.

c. Cure at a temperature of 74° ± 3°C (165° ± 5°F) for 120 ± 5 minutes and allow to cool. Finish specimens per ASTM D638.

d. Condition specimens at 23° ± 3° (73.4 ± 3°F) and relative humidity (R.H.) of 50 ± 5% for 4 hours prior to test.

e. The crosshead speed of testing shall be Speed A (0.005 in/min. ± 25%)

4.5.2.2 Bond strength. Bond strength shall be tested in accordance with ASTM D897 or equivalent subject to the following conditions:

a. The test specimen plates shall be SAE4130 steel.

b. The contact surfaces shall be a 32 microfinish or better.

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c. Specimens shall be cleaned by rinsing in methylene chloride and wiping with a clean cloth dampened in methyline chloride until no contaminants appear on the cloth. Air dry for a minimum of 30 minutes at a minimum temperature of 70°F.

d. Cure at a temperature of 74°F ± 3°C (165° ± 5°F) for 120 ± 5 minutes and allow to cool.

e. Condition specimens at 23° ± 3°C (73.4 ± 3°F) and R.H. of 50 ± 5% for 4 hours prior to test.

f. The crosshead speed shall be 0.05 in/min. ± 25%

4.5.2.3 Tensile shear. Tensile shear shall be tested in accordance with ASTM D1002 or equivalent subject to the same conditions as 4.5.2.2 above.

5. PACKAGING

5.1 Packaging, packing and marking.

5.1.1 Bulk materials. When bulk material is specified in the contract or order, materials shall be packaged, packed and marked in accordance with MIL-STD-1188 (commercial pack). Marking on individual containers of each material shall include the following:

a. Item nomenclature
b. Part or dwg. no.
c. Used on M753 Projectile
d. Manufacturer
e. Lot no. and date of manufacture
f. Contract or purchase order no.
g. Quantity inclosed

5.1.2 Individual packaging. When individual packaging is specified in the contract or order, materials shall be packaged, packed and marked in accordance with dwg. 9321334. Packing and packing marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The adhesive system covered by this specification is intended for use in assembly of insulation/bulkheads for M753, 8 Inch Projectiles.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Type of packaging (see 5.)
c. First article inspection, when required (see 3.3).

6.3 Test samples. Material samples shall be submitted for approval to Commander US Army Armament Research and Development Command, ATTN: DRDAR-LCA-OA, Dover, New Jersey 07801 or an approved laboratory.

6.4 Supersession data. This specification includes the requirements of AR-PD-10 dated 1 June 1978.

6.5 Drawings and specifications. Drawings and specifications listed in Section 2 of this specification under the heading US Army Armament Research and Development Command may also include dwgs. and specifications prepared and identified as Frankford Arsenal, Edgewood Arsenal, Rock Island Arsenal or Picatinny Arsenal dwgs. and specifications. Technical data originally prepared by these activities are now under the cognizance of ARRADCOM.

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MIL-A-63187 ADHESIVE, POLYAMIDE EPOXY M753 8 INCH PROJECTILE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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